# Spread of epidemic disease model

Epidemic disease affects a large no. of people within a community, population or region.

Epidemic of infectious disease have always been a subject of great concern for the welfare of community. Here, we introduce and discuss three simple differential equation models for the spread of epidemic.

1) The SI Model (Susceptibles - Infective)
2) The SIS Model (Susceptibles - Infective - Susceptibles)
3) The SIR Model (Susceptibles - Infective - Removed)

## Statement of the Problem:

Consider a spread by of epidemic through a single urban population. obtain a differential equation model so that we can predict what might happen with this kind of epidemic in the future.

## Formulation of the Model:

We divide the fixed total population (N) at any time t into three distinct groups.

→ Those who are infected and capable of spreading the disease to others. are known as Infecteds (I)

→ Those who have recovered and now have an immunity to the disease or died due to infection are known as Removeds (R)

→ Those who are still susceptible and are capable of catching the disease are known as Susceptibles (S)

Here time (t) is independent variable and two related sets of dependent variable are considered as a function of time.

First set:

→ S(t) is the number of Susceptibles individuals.

→ I(t) is the number of infected individuals.

→ R(t) is the number of recovered individuals.

## Second Set

$s(t) = \frac{S(t)}{N}$ is the susceptible fraction of the population

$i(t) = \frac{I(t)}{N}$ is the infected fraction of the population

$r(t) = \frac{R(t)}{N}$ is the recovered fraction of the population.

It may seem more natural to work with population counts, but some of our calculations will be simpler if we use the fractions instead. The two sets of dependent variables are proportional to each other, so either set will give us the same information about the progress of the epidemic.

Again, we have at each time $t$

$$s(t) + i(t) + r(t) = 1 \longrightarrow (1)$$

Following are the assumptions to be made

1) The time rate of change of $s(t)$ depends on the number already susceptible, the number of individuals already infected, and the amount of contact between the susceptibles and infected.

2) Suppose that each infected individual has a number $b$ of contact per day that are sufficient to spread the disease. Not all these contacts are with susceptibles individuals. If we assume a homogeneous mixing of the population, the fraction of these contacts that are with susceptibles is $s(t)$. Thus on an average, each infected individual generates $b \cdot s(t)$ new infected individuals per day. [With a large susceptible population and a relatively small infected population, we can ignore tricky counting situations as a single susceptible encountering more than one infected in a given day.

We also assume that a fixed fraction of k of the infected group will recover during any given day. (For eg: If the average duration of infection is three days, then on average, one-third of the currently infected population recovers each day).

The susceptible equation is

$$\frac{ds}{dt} = -b \cdot s(t) \cdot i(t) \longrightarrow (2)$$

The recovered equation is

$$\frac{dr}{dt} = k \cdot i(t) \longrightarrow (3)$$

The infected equation

$$\frac{ds}{dt} + \frac{di}{dt} + \frac{dr}{dt} = 0 \longrightarrow (4)$$

or

$$\frac{di}{dt} = -\frac{ds}{dt} - \frac{dr}{dt}$$

since $s + i + r = 1$

$$\frac{d}{dt}(s+i+r) = 0$$

$$\Rightarrow \frac{ds}{dt} + \frac{di}{dt} + \frac{dr}{dt} = 0.$$

## Model I: Susceptible Infected (SI) model.

Here, we assume that no person is recovered from the disease during the time of epidemic

$\therefore r(t) = 0$ at any time $t$

i.e $s + i = 1$.

Initially, the no. of infectives is $i_0$ (small) and that of susceptibles is $s_0$ (large)

i.e $i(0) = i_0$ and $s(0) = s_0 = 1 - i_0$ ($\because s_0 + i_0 = 1$)

From eqn (2), we get

$$\frac{ds}{dt} = -bs(1-s) \; ; \; s(0) = s_0 \longrightarrow (5)$$

Again $\frac{di}{dt} = -\frac{ds}{dt}$

$$\frac{di}{dt} = b.s.i = bi(1-i) \; ; \; i(0) = i_0 \longrightarrow (6)$$

Soln: Eqn (6) is solved by the method of separation of variables.

$$\therefore \frac{di}{i(1-i)} = b\,dt$$

$$\Rightarrow \frac{di}{i(1-i)} = \frac{A}{i} + \frac{B}{1-i} = \frac{A(1-i)+B(i)}{i(1-i)} = \frac{A+i(B-A)}{i(1-i)}$$

comparing the coefficients,

$$A=1 \text{ and } B-A=0$$

$$\Rightarrow B=1$$

$$\therefore \frac{1}{i} + \frac{1}{1-i} = b\,dt$$

On integrating, we get

$$\log i + \log(1-i)(-1) = bt + c$$

$$\therefore \log \frac{i}{1-i} = bt+c$$

$$\Rightarrow \frac{i}{1-i} = e^{bt+c} \Rightarrow \frac{i}{1-i} = A\,e^{bt}\,e$$

$$\Rightarrow \frac{i}{1-i} = A\,e^{bt} \quad [\because A = e^c]$$

$$\Rightarrow i = A\cdot e^{bt}(1-i)$$

Now, $i(0) = i_0$

$$\Rightarrow A = \frac{i_0}{1-i_0}$$

$$\therefore i = \frac{i_0}{1-i_0}\cdot e^{bt}(1-i)$$

$$\therefore i = A'(1-i) \quad \text{where } A' = \frac{i_0}{1-i_0}e^{bt}$$

$$i = A' - A'i$$

$$i(1+A') = A'$$

$$i = \frac{A'}{1+A'} \Rightarrow i(t) = \frac{(i_0/(1-i_0))\cdot e^{bt}}{1+(i_0/(1-i_0))\cdot e^{bt}}$$

$$= \frac{\frac{i_0}{1-i_0}\cdot e^{bt}}{\frac{i_0}{(1-i_0)}\cdot e^{bt}\left[\frac{1}{\frac{1-i_0}{i_0}e^{-bt}}+1\right]} = \frac{1}{\frac{1-i_0}{i_0}e^{-bt}+1}$$

$$\therefore i(t) = \frac{1}{\frac{1-i_0}{i_0} e^{-bt} + 1} \longrightarrow (7)$$

Similarly,

$$s(t) = 1 - i(t)$$

$$= 1 - \frac{1}{\frac{1-i_0}{i_0} \cdot e^{-bt} + 1} = \frac{\frac{1-i_0}{i_0} e^{-bt} + 1 - 1}{\frac{1-i_0}{i_0} \cdot e^{-bt} + 1}$$

$$= \frac{\frac{1-i_0}{i_0} e^{-bt}}{\frac{1-i_0}{i_0} e^{-bt} \left[1 + \frac{i_0}{1-i_0} e^{bt}\right]}$$

$$\therefore s(t) = \frac{1}{1 + \frac{i_0}{1-i_0} e^{bt}} \longrightarrow (8)$$

Thus, from eq$^n$ (7) and (8), it follows that $\lim_{t \to \infty} i(t) = 1$ and $\lim_{t \to \infty} s(t) = 0$

That is after a long time no individual remains susceptible and the whole population becomes infected as our assumption is that no person is removed (cured) from the disease.

## Model II: Susceptible Infected Susceptibles Model (SIS)

Here, we assume that an infected individual can be recover from the disease and again become susceptible at the rate k.

Thus, we have

$$\frac{ds}{dt} = -b \cdot s \cdot i + k \cdot i$$

$$\frac{ds}{dt} = -bs(1-s) + k(1-s)$$

$$\frac{ds}{dt} = (1-s)(-bs+k) \; ; \; s(0) = s_0 \longrightarrow (9)$$

$$\frac{ds}{dt}$$

Now, $\frac{di}{dt} = -\frac{ds}{dt}$

$\frac{di}{dt} = bsi - ki$

$\frac{di}{dt} = bi(a-i)$ where $a = 1 - \frac{k}{b}$ ; $i(0) = i_0$ → (10)

Now, solve eqn (10) by variable separable method

$\Rightarrow \frac{di}{i(a-i)} = b \cdot dt$

$\Rightarrow \frac{a\,di}{i(a-i)} = a \cdot b \cdot dt$

since
$bi(a-i) = bi\left(1 - \frac{k}{b} - i\right)$
$= bi\left(\frac{b-k-ib}{b}\right)$
$= bi\left(\frac{b(1-i)-k}{b}\right)$
$= bi\left(\frac{bs-k}{b}\right) = bi(s-k)$
$= bsi - ki$

$\Rightarrow \frac{a}{i(a-i)} = \frac{A}{i} + \frac{B}{a-i}$

$\Rightarrow \frac{a}{i(a-i)} = \frac{A(a-i) + Bi}{(a-i) \cdot i}$

$\therefore A = 1$ & $B - A = 0$
$\Rightarrow B = 1$

$\therefore \left[\frac{1}{i} + \frac{1}{a-i}\right] di = ab \cdot dt$

On, integrating we get

$\log i - \log(a-i) = abt + A'$

$\therefore \log \frac{i}{a-i} = abt + A'$

$\Rightarrow \frac{i}{a-i} = e^{abt} \cdot e^{A'}$

$\Rightarrow \frac{i}{a-i} = A \cdot e^{abt}$ where $A = e^{A'}$

Again $i(0) = i_0 \Rightarrow A = \frac{i_0}{a - i_0}$

$\therefore \frac{i}{a-i} = \frac{i_0}{a-i_0} \cdot e^{abt}$

$$\therefore i = \frac{i_0}{a - i_0} \cdot e^{abt} \cdot (a - i)$$

$$i = A'(a - i) \qquad \left[ A' = \frac{i_0}{a - i_0} \cdot e^{abt} \right]$$

$$\therefore i + A'i = A'a$$

$$\therefore i(1 + A') = A'a$$

$$\therefore i = \frac{A'a}{1 + A'} \Rightarrow i = \frac{\frac{i_0}{a - i_0} e^{abt} \cdot a}{1 + \frac{i_0 e^{abt}}{a - i_0}}$$

$$\Rightarrow i = \frac{i_0 e^{abt} \cdot a}{(a - i_0) + i_0 e^{abt}}$$

Now Divide N & D by $e^{abt}$

$$\therefore i = \frac{i_0 a}{(a - i_0) e^{-abt} + i_0}$$

$$\Rightarrow i(t) = \frac{a i_0}{i_0 + (a - i_0) e^{-abt}} \longrightarrow (11)$$

Similarly

$$s(t) = 1 - i(t)$$

$$= 1 - \frac{a i_0}{i_0 + (a - i_0) e^{-abt}}$$

$$= \frac{i_0 + (a - i_0) e^{-abt} - a i_0}{i_0 + (a - i_0) e^{-abt}}$$

$$s(t) = \frac{(1 - a) i_0 + (a - i_0) e^{-abt}}{i_0 + (a - i_0) e^{-abt}} \longrightarrow (12)$$

The soln of eqn (11) and (12) can be interpreted in two cases.

Case 1: When $a > 0$

$$1 - \frac{k}{b} > 0 \Rightarrow 1 > \frac{k}{b} \Rightarrow b > k$$

$\Rightarrow \lim_{t\to\infty} e^{-abt} = 0 \quad (\because e^{-\infty} = 0)$

$\therefore \lim_{t\to\infty} i(t) = a$ and $\lim_{t\to\infty} s(t) = 1-a$

Case 2: When $a < 0$

$1 - \frac{k}{b} < 0 \Rightarrow 1 < \frac{k}{b} \Rightarrow b < k$

$\Rightarrow \lim_{t\to\infty} e^{-abt} = \infty \quad [\because e^{\infty} = \infty]$

$\therefore \lim_{t\to\infty} i(t) = 0$ and $\lim_{t\to\infty} s(t) = 1$

## Model 3: Susceptible Infected removed (SIR) model

Here, we assume that an individual may be cured (removed) from the disease by death or getting permanent immunity.

From eq$^n$ (2), (3) and (4), we get

$$\frac{ds}{dt} = -bs.i \; ; \; s(0) = s_0 = 1 - i_0$$

$$\frac{di}{dt} = bsi - ki \; ; \; i(0) = i_0$$

$$\frac{dr}{dt} = ki \; ; \; r(0) = 0$$

By chain rule, we get

$$\frac{di}{ds} = \frac{di}{dt} \cdot \frac{dt}{ds}$$

$$= \frac{bsi - ki}{-bsi} = -1 + \frac{k}{bs}$$

$$\frac{di}{ds} = -1 + \frac{1}{c.s} \quad \left[\because c = \frac{b}{k} \rightarrow (13)\right]$$

where $c = \frac{b}{k}$ is known as contact number and defined as the number of close contacts per infected individual.

[∵ In most cases, the fraction $k$ of infecteds recovering in a given day can be estimated from observation of infected individuals. Specifically, $k$ is roughly the reciprocal of the number of days an individual is sick enough to infect other. For many contagious diseases, the infection time is approximately the same for the most infecteds and is known by ~~obsegth~~ observation.

$\therefore \frac{b}{k} = b \times \frac{1}{k}$ = no. of close contacts per day per infected × the no. of days infected

= no. of close contact per infected individual. ]

We call this ratio as the "contact number" and we write $c = \frac{b}{k}$. The contact number $c$ is combined characteristic of the population and of the disease. It measures the relative contagiousness of the disease, because it tells us indirectly how many of the contacts are close enough of the ~~contacts~~ to actually spread the disease.

Now, To find the sol$^n$ ~~of~~ of eq$^n$ (13) w.r.t $s$, we get

$$\frac{di}{ds} = -1 + \frac{1}{c \cdot s} \Rightarrow \frac{di}{ds} + 1 = \frac{1}{c \cdot s}$$

$$\Rightarrow di + ds = \frac{1}{c} \frac{1}{s} \cdot ds \quad \rightarrow (14)$$

$$\Rightarrow i + s = \frac{1}{c} \log s + q \Rightarrow \boxed{i + s - \frac{1}{c} \log s = q}$$

The term $i + s - \frac{1}{c} \log s$ is independent of time.

Now using the initial condition

$i(0) = i_0$, $s(0) = s_0$, $z(0) = 0$

$\therefore i_0 + s_0 - \frac{1}{c}\log s_0 = q.$

$$\boxed{i + s - \frac{1}{c}\log s = i_0 + s_0 - \frac{1}{c}\log s_0} \quad (15)$$

Suppose that long time after the onset of the epidemic, we have $i(\infty) \overset{=t}{.}$ approximate to zero and $s(\infty)$ has settled to its steady state value.

$\therefore i(\infty) = 0$ & $s(\infty) = s_0$

With the help of the SIR model, we can identify certain characteristics of the model like, the maximum number of infectives or the number of ~~is~~ recovered ~~be~~ individuals etc.